

# **Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?**

Leo Specks

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July 2027

Last Modified: Saturday 13<sup>th</sup> June, 2026 (14:04)  
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# 1 Test Chapter

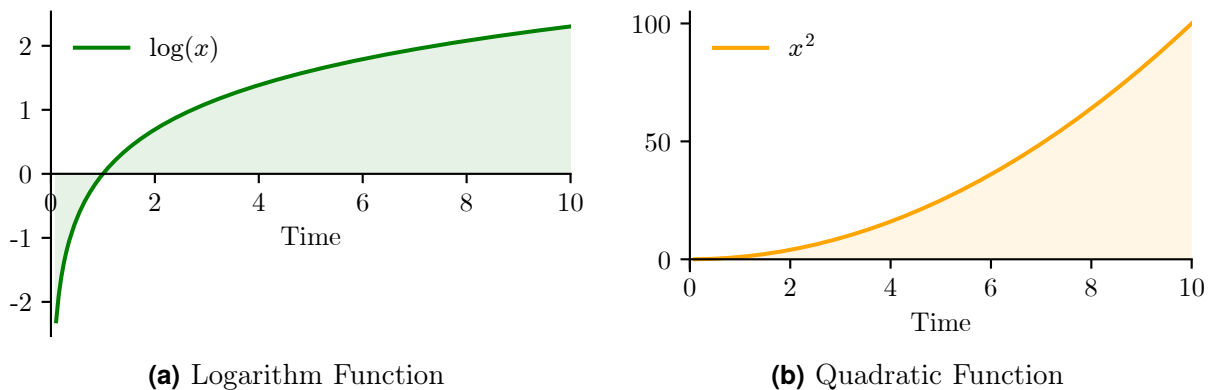
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**Figure 1**

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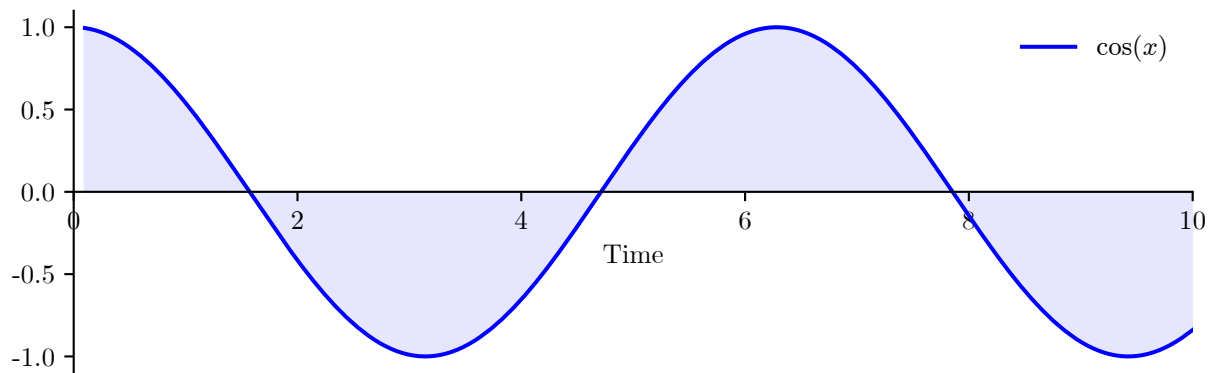


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**Figure 2**

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<sup>1</sup>This would be the first footnote.

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## 1.2 Citations

This is a `\cite{}`: [2], this a `\textcite{}`: Guidolin and Pedio [2], this a `\parencite{}`: [2], this a `\citeauthor{}`: Guidolin and Pedio, this a `\citeyear{}`: 2021, and this a `\parencite` with two references: [1, 2].

**Table 1**

Citation Styles with BibLaTeX

| Command                    | authoryear     | numeric           |
|----------------------------|----------------|-------------------|
| <code>\cite{}</code>       | Author, Year   | [index]           |
| <code>\textcite{}</code>   | Author (Year)  | Author [index]    |
| <code>\parencite{}</code>  | (Author, Year) | [index]           |
| <code>\citeauthor{}</code> | Author         | Author            |
| <code>\citeyear{}</code>   | Year           | Year (not linked) |

Use `\parencite{}` at the end of a statement and `\textcite{}` when referencing in the middle of a sentence.

## 1.3 Codeblock

The algorithm can be viewed in Codeblock 1. Nullam egestas viverra purus vitae varius. Sed maximus tempor vestibulum. Nunc tempor nulla justo, a posuere lorem tincidunt eget. Praesent dictum sed eros in pharetra. Vivamus nec dictum risus. Integer ac massa at magna rutrum pulvinar eget nec nunc. Curabitur tempor at lorem vitae venenatis. Suspendisse auctor posuere iaculis. Suspendisse egestas lobortis diam. Integer mattis.

**Code 1**

Example Codeblock

```
# KPI OVERVIEW
kpis = pd.DataFrame({
    "Ann. Return (approx)": ann_return_simple,
    "Ann. Return (compounding)": ann_return,
    "Ann. Volatility": ann_vol,
    "Sharpe": sharpe,
    "CAGR": cagr,
    "Max Drawdown": max_dd
})

kpis.T
```

## **2 Introduction**

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### **10.3 Future Research Directions**

---

## References

- [1] Marta Giampietro, Massimo Guidolin, and Manuela Pedio. “Estimating Stochastic Discount Factor Models with Hidden Regimes: Applications to Commodity Pricing”. In: *European Journal of Operational Research* 265.2 (Mar. 1, 2018), pp. 685–702. ISSN: 0377-2217. DOI: 10.1016/j.ejor.2017.07.045.
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- [3] Benjamin R. Auer. “Have Trend-Following Signals in Commodity Futures Markets Become Less Reliable in Recent Years?” In: *Financial Markets and Portfolio Management* 35.4 (Dec. 1, 2021), pp. 533–553. ISSN: 2373-8529. DOI: 10.1007/s11408-021-00385-5.
- [4] Massimo Guidolin and Serena Ionta. “Predicting Commodity Returns with Climate Variables: Statistical Loss Functions vs. Economic Value”. In: *Economics Letters* 265 (June 1, 2026), p. 113028. ISSN: 0165-1765. DOI: 10.1016/j.econlet.2026.113028.

\*not cited yet